

Que 1) An image scan center needs to store CAD images whose resolution is 1024×1024 , with 64 gray levels. A total of 100 layers are present.

A. How much storage is required in GB.

- Image size = $1024 * 1024 * 6 * 100$ bits
 $= 629145600$ bits

- Note: 1 Byte = 8bits, KB = 10^3 , MB = 10^6 , GB = 10^9

- Image size = $\frac{629145600}{10^9} = 0.629145600$ GB

B. Suppose this image is transferred via medium having a transfer rate of 128 K bits/second in the form of packets. The size of a packet is 100 bits and each packet is with a start and a stop flag of 14 bits each. Compute the time required to transfer the image (in hrs).

- Image size = $1024*1024*6*100$ bits
- Total size for transfer = $1024*1024*6*(100+14+14)$
- Time = $\frac{1024*1024*6*(100+28)}{128000}$
= $1024*1024*6/1000$ sec
= 6291.456 sec
= 1.747 hrs

Que 2) A common measure of transmission for digital data is the baud rate, defined as symbols (bits in our case) per second. As a minimum, transmission is accomplished in packets consisting of 14 start bits, a byte of information and 14 stop bits. Using these facts, answer the following:

- (i) How many seconds would it take to transmit a sequence of 200 images of size 1280 X 960 pixels with 256 intensity levels using a 3 M-baud modem? (This is a representative medium speed for digital subscriber line residential medium.)
- (ii) What would be the time using a 30 G-baud modem?

- Total size for transfer = $1280*960*8*(14+8+14)$
- For 200 Images = $1280*960*8*(14+8+14) \times 200$

$$(i) \text{ Time} = \frac{1280*960*8*(14+8+14) \times 200}{3 \times 10^6} \text{ sec}$$

$$(ii) \text{ Time} = \frac{1280*960*8*(14+8+14) \times 200}{3 \times 10^9} \text{ sec}$$